

Trainer Debrief — Lab 13

Nested and Destructive Gauge Studies (Black Belt Only)

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Design a valid MSA when parts cannot be measured repeatedly (A4, K2).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-13-nested-destructive-gauge.xlsx

A crossed Gage R&R is impossible here: the test destroys the part, so no operator can repeat another's measurement. Use a NESTED design and treat within-batch specimen variation as the repeatability estimate — valid only if the batch is homogeneous. Nested ANOVA gives measurement variation ~21% of study variation (marginal); batch-to-batch dominates, which is what you want.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

You can explain why a crossed study is invalid for your measurement, your batches are justified as homogeneous, and your report states the repeatability/part-variation confounding explicitly.

Where this sits in the capstone

BLACK BELT ONLY. The crossed Gage R&R of Lab 11 assumes every operator can measure every part repeatedly. Destructive tests (tensile, burst, seal strength, chemical assay) break that assumption entirely — measuring consumes the part. The nested design is the correct answer, and using a crossed study here produces confident nonsense. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27
Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30

If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

Certified Lean Six Sigma Black Belt (CLSSBB) Training · TGS-2024051900 · Version v4 · © 2026 Tertiary Infotech Academy Pte Ltd · TRAINER ONLY